

# SQL Assignment — 50 Questions

Beginner to Intermediate • Questions Only • No Answer Key

**Instructions:** Write SQL queries for all questions. Use the database tables provided below. Show your query and, where required, the resulting output.

## Database Schema and Sample Data

**Table: employees**

| employee_id | employee_name | department_id | manager_id | salary | city      | hire_date  |
|-------------|---------------|---------------|------------|--------|-----------|------------|
| 101         | Alice         | 10            | 201        | 60000  | Hyderabad | 2022-01-15 |
| 102         | Bob           | 20            | 202        | 55000  | Bangalore | 2021-06-20 |
| 103         | Charlie       | 10            | 201        | 70000  | Hyderabad | 2020-03-10 |
| 104         | David         | 30            | 203        | 50000  | Mumbai    | 2023-02-18 |
| 105         | Emma          | 40            | 204        | 65000  | Delhi     | 2022-11-05 |
| 106         | Frank         | NULL          | 201        | 45000  | Hyderabad | 2024-01-10 |
| 107         | Grace         | 20            | 202        | 58000  | Bangalore | 2023-07-12 |
| 108         | Henry         | 30            | 203        | 52000  | Mumbai    | 2021-09-25 |
| 109         | Irene         | 10            | 201        | 72000  | Hyderabad | 2019-05-14 |
| 110         | Jack          | 50            | NULL       | 48000  | Chennai   | 2024-04-01 |

**Table: departments**

| department_id | department_name | location  | budget |
|---------------|-----------------|-----------|--------|
| 10            | Data Science    | Hyderabad | 500000 |
| 20            | HR              | Bangalore | 300000 |
| 30            | Finance         | Mumbai    | 400000 |
| 40            | Sales           | Delhi     | 450000 |
| 50            | Marketing       | Chennai   | 350000 |
| 60            | IT              | Pune      | 600000 |

**Table: projects**

| project_id | project_name       | department_id | budget |
|------------|--------------------|---------------|--------|
| 1          | AI Platform        | 10            | 250000 |
| 2          | Recruitment System | 20            | 150000 |
| 3          | Fraud Detection    | 30            | 200000 |
| 4          | Sales Dashboard    | 40            | 180000 |
| 5          | Marketing Campaign | 50            | 120000 |
| 6          | Cloud Migration    | 60            | 300000 |

**Note:** NULL means no value. The students table is to be created in Question 29.

## Section A — Basic SQL

- Q1. Display all employees.
- Q2. Display only employee name and salary.
- Q3. Display employees who earn more than ■60,000.
- Q4. Display employees who live in Hyderabad.
- Q5. Display employees whose salary is between ■50,000 and ■70,000.
- Q6. Display employees who live in either Hyderabad or Bangalore.
- Q7. Display employees whose names start with the letter 'A'.
- Q8. Display employees whose names end with the letter 'e'.
- Q9. Display all employees sorted by salary from highest to lowest.
- Q10. Display the top 3 highest-paid employees.

## Section B — Aggregate Functions

- Q11. Find the total number of employees.
- Q12. Find the total salary of all employees.
- Q13. Find the average employee salary.
- Q14. Find the highest salary.
- Q15. Find the lowest salary.
- Q16. Find the number of employees living in Hyderabad.
- Q17. Find the total salary paid to employees living in Mumbai.
- Q18. Find the average salary of employees living in Bangalore.

## Section C — GROUP BY and HAVING

- Q19. Find the number of employees in each department.
- Q20. Find the average salary in each department.
- Q21. Find the maximum salary in each department.
- Q22. Find the minimum salary in each department.
- Q23. Find the total salary paid by each department.
- Q24. Find the number of employees in each city.
- Q25. Find the average salary for each city.
- Q26. Find cities having more than 2 employees.
- Q27. Find departments where the average salary is greater than ■60,000.
- Q28. Find departments where the total salary is greater than ■150,000.

## Section D — Constraints

**Q29.** Create the students table using the following constraints: student\_id PRIMARY KEY, student\_name NOT NULL, email UNIQUE, age CHECK (age >= 18), and city DEFAULT 'Hyderabad'.

**Q30.** Insert a valid student record into the students table.

**Q31.** Attempt to insert two students with the same student\_id. Identify the constraint that is violated.

**Q32.** Attempt to insert a student without providing student\_name. Identify what happens.

**Q33.** Attempt to insert a student whose age is 15. Identify the constraint that is violated.

**Q34.** Insert a student without specifying city. Verify the value assigned by the DEFAULT constraint.

## Section E — JOINS

**Q35.** Display employee name and department name.

**Q36.** Display employee name, salary, department name, and department location.

**Q37.** Display all employees, including employees who do not belong to any department.

**Q38.** Display all departments, including departments that have no employees.

**Q39.** Find employees working in the Data Science department.

**Q40.** Find employees working in departments located in Hyderabad.

**Q41.** Find the number of employees in each department using JOIN and GROUP BY. Include departments with zero employees.

**Q42.** Find the average salary for each department using JOIN and GROUP BY.

## Section F — Subqueries

**Q43.** Find employees whose salary is greater than the overall average salary.

**Q44.** Find the employee with the highest salary using a subquery.

**Q45.** Find employees who earn the same salary as Alice.

**Q46.** Find employees who earn more than Alice.

**Q47.** Find employees who earn more than the average salary of their own department.

## Section G — Mixed SQL

**Q48.** Find the department with the highest average salary.

**Q49.** Display each department with department name, employee count, average salary, maximum salary, minimum salary, and total salary. Include departments with zero employees.

**Q50.** Find employees who satisfy all conditions: salary greater than ₹55,000; city is Hyderabad or Bangalore; employee belongs to a valid department; and employee salary is greater than the average salary of their department.